

SOLAR SECTOR At The Crossroads



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Energy demand in India is far outstripping the supply. Solar energy offers a huge opportunity to bridge this widening gap between demand and supply. However, there are concerns over the solar industry's readiness against the backdrop of a sleuth of changes in the business environment like continuous tariff falls, the slow pace of new tender announcements and the introduction of GST. Let's delve deeper to get the complete picture.

India grabs the power baton

A survey by Wood Mackenzie on the global energy markets, which focuses on the upcoming market shifts, concludes that global energy demand will keep rising through 2035. However, the nature of the energy mix, demand centres and key players will change significantly. The report also predicts that energy demand in India will grow the fastest, at an average rate of 3 per cent annually from now to 2035, by which time the country's population will outgrow even China's. Renewable energy sources will surge from 22 per cent to 54 per cent of the total installed capacity, while oil consumption will rise by 80 per cent. According to an Indian Brand Equity Foundation (IBEF) report, India ranks third among 40 countries in Ernst and

Young's Renewable Energy Country Attractiveness Index.

With a production of 1,400,800GWh in 2016, India is the third largest producer as well as the third largest consumer of electricity in the world. Although power generation has grown manifold since Independence, demand has been even higher due to accelerating economic activity. India has a diversified power sector with energy sources ranging from the conventional coal, natural gas, oil, hydro and nuclear power to renewable sources such as

wind, solar and biological waste. According to data available from the Ministry of Power (Table), renewable energy sources accounted for around 17.4 per cent of India's total installed power capacity in March 2017.

Solar rising faster

Solar power has quickly become a sizeable part of India's energy mix. According to GTM Research and a Bridge To India (BTI) report, the total solar PV capacity installed globally had surpassed 300GW by the end of 2016. Of this, 77GW was added in 2016—a year-on-year growth rate of 34 per cent. China led with 34.5GW, followed by the USA (14.5GW) and Japan (10.2GW), with India (5GW) in fourth place.

In 2017, India is expected to continue its rapid growth in solar power. With 8.8GW of projected capacity addition (a growth of 76 per cent over 2016), the country is set to become the third largest PV market in 2017, overtaking Japan (Figs 1 and 2). The GTM report also mentioned that, as of March 31, 2017, India had installed 12.2GW of utility-scale solar PV capacity. Till date, the southern states of Tamil Nadu, Andhra Pradesh and Telangana have taken the lead in solar installation projects (Fig. 3). This year, approximately 60 per cent of new solar capacity addition is expected to come from Telangana, Andhra Pradesh and Karnataka.

Tariff at stake

The wide-scale adoption of solar energy, coupled with the sharp decline in solar module prices in the global market by almost 30 per cent (a trend that is likely to continue next year also) due to a supply glut, has resulted in an abrupt fall of solar power tariff in India. In the last one year, the Indian solar sector experienced almost 50 per cent tariff reduction, with 25 per cent drop happening in just the last three months.

In May this year, India's solar power tariffs fell to a new all-time low of 3.8 US cents (approx. ₹ 2.40) per unit during the

INDIA'S ENERGY USE PATTERN IN 2016

Fuel	MW	% of Total
Total Thermal	220,576	67.0%
Coal	194,553	59.1%
Gas	25,185	7.6%
Oil	838	0.3%
Hydro (Renewable)	44,614	13.6%
Nuclear	6780	2.1%
RES* (MNRE)	57,260	17.4%
Total	329,231	100%

*Installed capacity with respect to RES (MNRE) as on 31.03.2017
RES (renewable energy sources) includes small hydro projects, biomass gasifiers, biomass power, urban industrial waste-based power, solar and wind energy.
Source: Ministry of Power